

Vector product

Basic skills

$$\text{Q1. a)} \quad \underbrace{\begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}}_{\underline{a}} \times \underbrace{\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}}_{\underline{b}} = \begin{pmatrix} 2 \times 2 - 1 \times 1 \\ 5 \times 2 - 1 \times 3 \\ 5 \times 1 - 2 \times 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ -1 \end{pmatrix}$$

$\begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_1 b_3 - a_3 b_1 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$

$$\text{b)} \quad \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \times 0 - 1 \times 1 \\ 5 \times 0 - 1 \times 2 \\ 5 \times 1 - -2 \times 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 9 \end{pmatrix}$$

$$\text{c)} \quad \begin{pmatrix} 6 \\ 10 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 10 \times -1 - 2 \times -2 \\ 6 \times -1 - 2 \times 3 \\ 6 \times -2 - 10 \times 3 \end{pmatrix} = \begin{pmatrix} -6 \\ -12 \\ -42 \end{pmatrix}$$

Q2.

$$\begin{vmatrix} 3 & -2 & -1 \\ 4 & 3 & 5 \\ -3 & 1 & -2 \end{vmatrix} = 3 \begin{vmatrix} 3 & 5 \\ 1 & -2 \end{vmatrix} - -2 \begin{vmatrix} 4 & 5 \\ -3 & -2 \end{vmatrix} - 1 \begin{vmatrix} 4 & 3 \\ -3 & 1 \end{vmatrix}$$
$$= 3(-6-5) + 2(-8+15) - (4+9)$$
$$= -33 + 14 - 13 = -32$$

Q3.

a) $\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \leftarrow \underline{a} \rightarrow \\ \leftarrow \underline{b} \rightarrow \end{vmatrix}$ so $\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 1 & -2 \\ 2 & 1 & 4 \end{vmatrix}$

b) $= \underline{i} \begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix} - \underline{j} \begin{vmatrix} 0 & -2 \\ 2 & 4 \end{vmatrix} + \underline{k} \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix}$
 $= \underline{i} (4+2) - \underline{j} (0+4) + \underline{k} (0-2)$
 $= 6\underline{i} - 4\underline{j} - 2\underline{k}$

Competency skills

Q4. $\underline{a} = \begin{pmatrix} -2 \\ 1 \\ -3 \end{pmatrix}, \underline{b} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}$

a) As $\underline{b} = -2\underline{a}$ parallel

b) $\underline{a} \times \underline{b} = \begin{pmatrix} 1 \times 6 - 3 \times -2 \\ -3 \times 4 - -2 \times 6 \\ -2 \times -2 - 1 \times 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \underline{0}$

formula
or
determinant
trick

c) $\underline{a}, \underline{b}$ Parallel $\Rightarrow \underline{a} \times \underline{b} = \underline{0}$

$\underline{a}, \underline{b}$ Perpendicular $\Rightarrow \underline{a} \cdot \underline{b} = 0$

Q5. $\underline{a} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}, \underline{b} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$

$$\underline{a} \times \underline{b} = \begin{pmatrix} 3 \times 1 - 5 \times 2 \\ 5 \times 4 - 1 \times 1 \\ 1 \times 2 - 3 \times 4 \end{pmatrix} = \begin{pmatrix} -13 \\ 21 \\ -10 \end{pmatrix}$$

$$(\underline{a} \times \underline{b}) \cdot \underline{a} = \begin{pmatrix} -13 \\ 21 \\ -10 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} = -13 + 63 - 50 = 0$$

$$(\underline{a} \times \underline{b}) \cdot \underline{b} = \begin{pmatrix} -13 \\ 21 \\ -10 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = -52 + 42 + 10 = 0$$

Hence $(\underline{a} \times \underline{b}) \cdot \underline{a} = 0 = (\underline{a} \times \underline{b}) \cdot \underline{b}$

meaning $\underline{a} \times \underline{b}$ is perpendicular to both $\underline{a}$ and $\underline{b}$.

Q6. $\theta = 30^\circ$ $\underline{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ direction of Product, needs unitising

need

$$\underline{\hat{n}} = \frac{\underline{n}}{|\underline{n}|} = \frac{1}{\sqrt{3^2 + (-1)^2 + 2^2}} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{14}} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

Then as $\underline{a}, \underline{b}$ unit vectors $|\underline{a}| = |\underline{b}| = 1$

so $\underline{a} \times \underline{b} = |\underline{a}| |\underline{b}| \sin \theta \underline{\hat{n}}$

$$= 1 \times 1 \times \sin 30^\circ \times \frac{1}{\sqrt{14}} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

$$= \frac{1}{2\sqrt{14}} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

Advanced skills

Q7. TO show $\underline{a} \times \underline{b} = -\underline{b} \times \underline{a}$

compute $\underline{b} \times \underline{a}$

so $\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \times \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} b_2 a_3 - b_3 a_2 \\ b_3 a_1 - b_1 a_3 \\ b_1 a_2 - b_2 a_1 \end{pmatrix} = - \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$

could use determinant $= -\underline{a} \times \underline{b}$

so $\underline{b} \times \underline{a} = -\underline{a} \times \underline{b} \Rightarrow \underline{a} \times \underline{b} = -\underline{b} \times \underline{a}$

Q8. To show $|\underline{a} \times \underline{b}|^2 = |\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2$

compute $|\underline{a} \times \underline{b}|^2$

$$\begin{aligned}
 & \begin{vmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{vmatrix}^2 \\
 &= (a_2 b_3 - a_3 b_2)^2 + (a_3 b_1 - a_1 b_3)^2 \\
 &\quad + (a_1 b_2 - a_2 b_1)^2 \\
 &= a_2^2 b_3^2 - 2a_2 a_3 b_2 b_3 + a_3^2 b_2^2 \\
 &\quad + a_3^2 b_1^2 - 2a_1 a_3 b_1 b_3 + a_1^2 b_3^2 \\
 &\quad + a_1^2 b_2^2 - 2a_1 a_2 b_1 b_2 + a_2^2 b_1^2 \\
 &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \\
 &\quad - a_1^2 b_1^2 - a_2^2 b_2^2 - a_3^2 b_3^2 \\
 &\quad - 2(a_2 a_3 b_2 b_3 + a_1 a_3 b_1 b_3 + a_1 a_2 b_1 b_2) \\
 &= |\underline{a}|^2 |\underline{b}|^2 - (a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \\
 &= |\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2
 \end{aligned}$$

Q9. vector Product is perpendicular to both vectors, so in 2D it is not possible (unless parallel and then vector Product is 0)

Q10. $\underline{a} = \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}, \underline{b} = \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix}$

a) $\underline{a} \times \underline{b}$ is perpendicular to both

$$\underline{a} \times \underline{b} = \begin{pmatrix} 5 \times 0 - 1 \times -2 \\ 1 \times 5 - 5 \times 0 \\ 5 \times -2 - 5 \times 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ -35 \end{pmatrix}$$

For unit length $\frac{\begin{pmatrix} 2 \\ 5 \\ -35 \end{pmatrix}}{\sqrt{2^2 + 5^2 + 35^2}} = \frac{1}{\sqrt{1254}} \begin{pmatrix} 2 \\ 5 \\ -35 \end{pmatrix}$

b) As points on plane perpendicular to direction off plane

doesn't have to be unit

$$(\underline{r} - \underline{a}) \cdot \underline{n} = 0 \Rightarrow \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ 5 \\ -35 \end{pmatrix} = 0$$
$$\Rightarrow 2x + 5y - 35z = 0$$